

## FUZZY LOGIC

- A branch of AI that can deal with **vague or imprecise** data
- Traditional Boolean Logic has 2 outcomes – True or False (1 or 0)
- Fuzzy Logic has values between true and false (half true, almost true etc.)

## FUZZY LOGIC

- Fuzzy Logic allows us to write programs using Linguistic Terms (i.e. Natural Language Words)

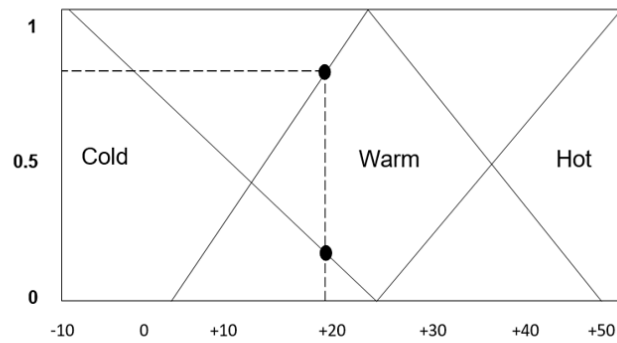
When does the speed of a car change from “Fast” to “Okay” ?

There is **NO** precise answer. Fuzzy Logic Deals with this by using **Fuzzy Sets**. A given input may have membership in more than one set.

# Stages of Fuzzy Logic Processing

## 1. Fuzzification

Fuzzification is the process of converting **precise input values taken from sensors** to **linguistic terms** (i.e. natural language terms). For example it may map the input temperature of **-10°** Celsius to linguistic term **"very cold"**.



Consider an input temperature of 20 degrees Celsius (shown by dashed line). This has a degree of **0.3** in the cold set and a degree of **0.8** in the warm set. Fuzzification would map this to a linguistic value such as "Slightly warm".

# Stages of Fuzzy Logic Processing Cont.

## 2. Rule based inference

The fuzzified input stored in a linguistic variable is then used in the **rule-based inference system**. This system uses previously defined **linguistic rules** to generate a fuzzy response.  
Example:

```
IF temperature = "very cold" THEN  
    Command = "set heater to a high level"  
END IF
```

## 3. Defuzzification

The fuzzy response generated by inference rules is then defuzzified in the final process. This process will convert the fuzzy output command to a crisp value. For example, the command **"set heater to a high level"** might map to level **8** of a **10-level heater**.

# FUZZY LOGIC APPLICATIONS

Fuzzy logic is often used to design real time control systems such as **subway train control systems**, **climate control AC systems**, **fully automatic washing machines**, **auto focus cameras** and **robotic vehicles**.

One of the first major successes of Fuzzy Logic was the design of a subway control system by **Hitachi** for their Sendai line.

## Key Aspects of the Fuzzy Logic System

- **Human-Like Control:** Developed by Hitachi engineers, including Dr. Seiji Yasunobu, the system was designed to emulate the decision-making process of a skilled human operator rather than relying on strict, fixed, or traditional binary algorithms.
- **Performance Improvements:**
  - **Smoothness:** The system provides smoother starts and stops compared to conventional, non-fuzzy automated systems.
  - **Stop Precision:** The accuracy of stopping the train at stations was increased by a factor of 3 (3X).
  - **Energy Efficiency:** The system is approximately 10% more energy-efficient than traditional methods.
- **Operation:** The controller uses "if-then" rules to assess inputs like station-to-station distance, target time, and current speed to determine the appropriate acceleration or braking force, acting on "degrees" of need rather than just ON/OFF.

## FUZZY LOGIC APPLICATIONS Cont.

- Fuzzy logic is used in **web searching**
- People often give imprecise search keys with incorrect spelling
- Fuzzy Logic can map it into nearest best match

# FUZZY SEARCHING - STRING MATCHING

- The **Levenshtein Distance** is an algorithm for measuring the closeness of one string to another
- It's a metric named after **Vladimir Levenshtein**, who originally considered it in 1965 to measure the difference between two sequences of words. We can use it to discover the minimum number of edits that you need to do to change a one-word sequence into the other.
- Example: To turn "BAT" into "TAB" 2 edits are required, to turn "BAT" into "BAG" only 1 edit is required